Calculation and Verification of Frequency Deviation and Modulation Index of FM Signals Using MATLAB Simulation

Aim: To calculate and verify the frequency deviation and modulation index of a frequency-modulated (FM) signal using MATLAB simulation.

To generate an FM signal using MATLAB.
To calculate its frequency deviation and modulation index.
To verify the relationship between these values through simulation.

CODE :

```
clc;
clear;
close all;

% Message signal parameters
Am = 2;            % Message amplitude (V)
fm = 500;          % Message frequency (Hz)

% Carrier signal parameters
Ac = 1;            % Carrier amplitude (V)
fc = 10000;        % Carrier frequency (Hz)

% FM parameters
kf = 250;          % Frequency sensitivity (Hz/V)
fs = 100000;       % Sampling frequency (Hz)
T = 0.01;          % Simulation duration (s)
```

```
% Time vector
t = 0:1/fs:T;

% Generate message signal
m = Am*cos(2*pi*fm*t);

% Calculate frequency deviation and modulation index
delta_f = kf*Am;
beta = delta_f/fm;

% Generate FM signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

% Display results
fprintf('Frequency Deviation = %.2f Hz\n', delta_f);
fprintf('FM Modulation Index = %.2f\n', beta);

% Verify modulation index
beta_verify = delta_f/fm;
fprintf('Verified Modulation Index = %.2f\n', beta_verify);

if abs(beta - beta_verify) < 1e-6
   fprintf('Verification Successful\n');
else
   fprintf('Verification Failed\n');
```

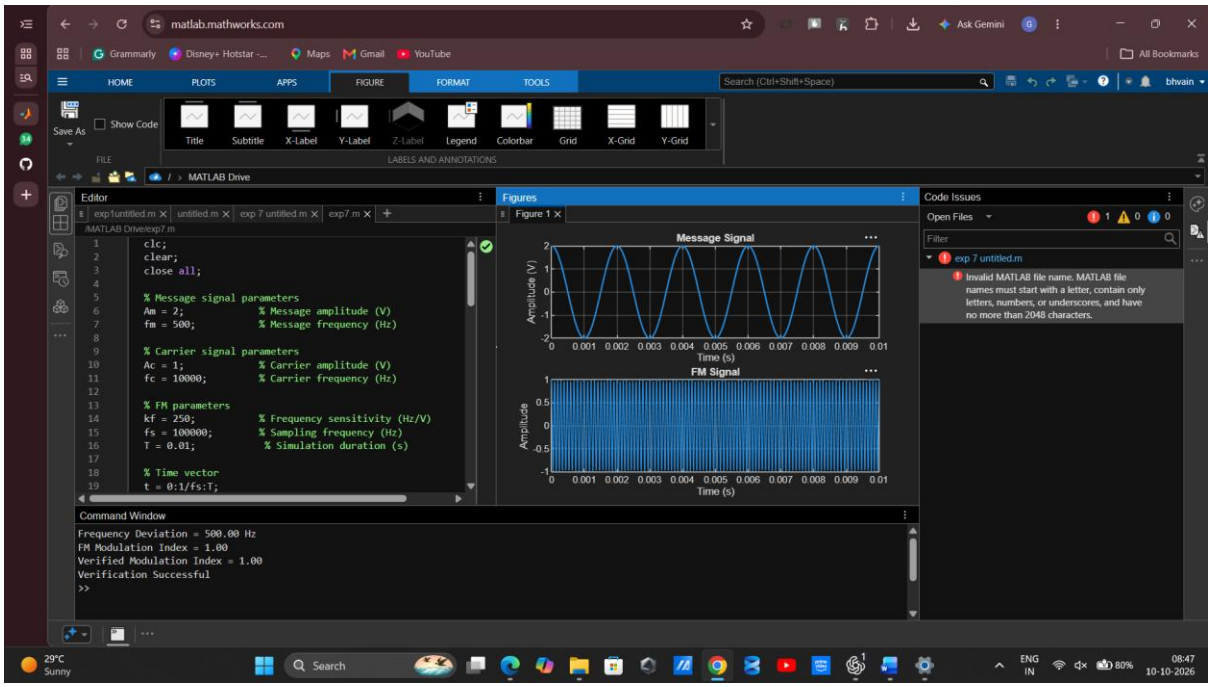
```
end

% Plot message and FM signals
figure;

subplot(2,1,1);
plot(t, m, 'LineWidth', 1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');

subplot(2,1,2);
plot(t, fm_signal, 'LineWidth', 1);
grid on;
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude');
```

OUTPUT :



RESULT:

The FM signal was generated successfully using MATLAB.
The frequency deviation was calculated as **500 Hz**.
The modulation index was **1**, and verification was successful.